CANFD bus transport layer

This chapter defines the CANFD bus based transport layer of DroneCAN.

DroneCAN on CANFD

DroneCAN operates on CANFD with a small number of changes from operation on bxCAN. The changes are:

- frames can be up to 64 bytes in length, as opposed to 8 byte frames for bxCAN
- the TAO (tail array optimisation) technique is never used in the encoding at any level
- the DLC is quantised as described below

Bus rates

CANFD implementations will usually use bit rate switching (BRS) with a nominal bit rate of 1MBps, with payload switching to 1, 2, 4, 5 or 8 MBit depending on hardware capabilities

DLC Encoding

To accomodate the limited range of the DLC field a mapping is used for data lengths of over 8 bytes.

The mapping is

```
# Data Length Code      9  10  11  12  13  14  15
# Number of data bytes 12  16  20  24  32  48  64
```

Compiler support

It is highly recommended that implementors use the dronecan_dsdlc compiler for building compliant C and C++ bindings for DroneCAN that work with both bxCAN and CANFD

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